Decision Matrix

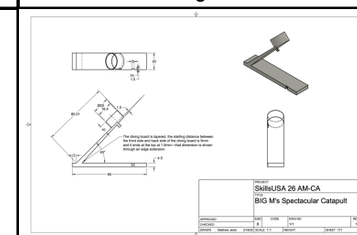
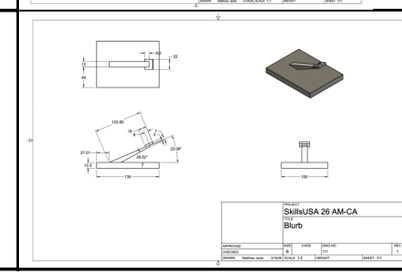
Step 1: Select criteria needed to solve the problem provided by SkillsUSA

Step 2: Rank the criteria using a scale from 1-5. Keep in mind what the criteria that the judges are looking for.

Step 3: Select the design to print and test. This is determined by the design that scores the highest on the rubric you created.

Team Name: **Momri**

Catapult Design Images

	Durability	**Theortical Functionality**	**Affordability**	**DfAM Optimization**	**Post-Processing**	**Legality**	5 max # of Points	Designer	Insert Fusion Designs:
BIG M'S SPECTACULAR DESIGN	3 — in simulaton, Fusion concluded it might permenantly deform.	4 — in simulation, Fusion concluded it would bend.	5 — in slicing, the design uses less then 50 cents worth of material and takes 6 hours to print.	5 — the print is able to be printed on its side without supports to utilize layer strength.	5 — the design lacks the need for support with DfAM optimization and is unified in one component.	5 — it is within dimensions and utilizes only 3D printed materials.	27	Matthew Jeide	
Blurb	5 — in simulation, Fusion concluded it would not deform.	3 — in simulation, Fusion concluded it would hardly bend, making it difficult to perform intended functionality.	2 — in slicing, the design uses more than 2 dollars worth of material and takes more than 18 hours to print.	2 — this design is not able to printed on its side without supports to utilize layer strength. The design is uncessarily large.	3 — the design requires supports to print properly, but it is unified in one component.	5 — it is within dimensions and utilizes only 3D printed materials.	20	Omri Lavi	

Describe the criteria that the judges might be looking for in your design. Select six criteria and describe below.

Criteria 1:	Durability — is the design durable and potentially capable of multiple launches and or rough handling?
Criteria 2:	Theortical Functionality — the effectiveness of the design.
Criteria 3:	Affordability — is the design effectively designed in terms of filament, time to print, and price?
Criteria 4:	DfAM Optimization — is the design optimally designed for additive manufacturing and FDM?
Criteria 5:	Post-processing — how many parts are printed and or supports needed? Fewer is better.
Criteria 6:	Legality — does the design meet the requirements outlined by the competition rules?